

Data Collection and Preprocessing Phase

Date	21-09-2026
Team ID	MCA
Project Title	Strategic Employee Attrition Intelligence
Maximum Marks	10 Marks

Data Exploration and Preprocessing Template

Identifies data sources, assesses quality issues like missing values and duplicates, and implements resolution plans to ensure accurate and reliable analysis.

Section	Description
Data Overview	5,000 employee records with 29 attributes covering demographics (age, gender, marital status, education level), job details (department, job role, job level), compensation (monthly income, stock option level, salary hike %), tenure (years at company, total working years, years since promotion, prior companies), workload (commute minutes, overtime hours/week, business travel days/year, training hours), wellbeing/engagement (manager support, burnout, engagement, work-life balance scores), performance (performance rating, last review score), AI adoption (uses_ai_tools_at_work, perceived_ai_job_risk), and the target variable attrition (Yes/No) — 895 employees (17.9%) have attrition = Yes.
Data Cleaning	Checked all 29 columns for missing values and imputed or removed them where appropriate (e.g. monthly_income, burnout_score, engagement_score); removed duplicate employee_id records; standardized inconsistent category labels in department, job_role, work_mode, and marital_status; validated that attrition and uses_ai_tools_at_work contain only valid Yes/No and True/False values.
Data Transformation	Used Power Query to filter out invalid or blank rows, sort records by department and job_role, pivot data to compute segment-level attrition rates by department, job role, and work mode, and add calculated columns such as tenure bands,

	income bands, and a burnout/engagement risk flag.
Data Type Conversion	Set employee_id to text; numeric fields (age, monthly_income, salary_hike_pct, years_at_company, total_working_years, years_since_promotion, num_prior_companies, commute_minutes, overtime_hours_per_week, business_travel_days_per_year, training_hours, manager_support_score, burnout_score, engagement_score, work_life_balance_score, performance_rating, last_review_score) to whole number/decimal; uses_ai_tools_at_work to True/False; and categorical fields (gender, marital_status, education_level, department, job_role, job_level, work_mode, perceived_ai_job_risk, attrition) to text categories.
Column Splitting and Merging	No columns required splitting; department and job_role were kept separate to support drill-down analysis, and a merged “Department – Job Role” column was created for combined segment-level visuals.
Data Modeling	Defined a single Employee fact table keyed on employee_id, with relationships to lookup tables for Department, Job Role, and Work Mode; built DAX measures for Attrition Rate (%), Average Burnout Score, Average Engagement Score, Average Tenure, and Average Salary Hike %, each segmented by department, job role, and work mode.
Save Processed Data	Saved the cleaned and transformed dataset within the Power BI data model (.pbix) and exported a backup cleaned CSV for reuse in future analysis and reporting.